

GOOGLE CLOUD AI BUILDER CUP 2026

GreenLedger

An auditable ESG disclosure agent for JAPAC SMEs.
Messy bills in. Cited IFRS S2 draft out.

Track: Sustainability & Social Impact · Team GreenLedger

Judging: Technical 40% · Impact 25% · Innovation 25% · UX 10%

Climate disclosure is now a supply-chain tax on SMEs

- SGX, HKEX, SSBJ and AASB S2 push IFRS S2-aligned reporting from FY2025 — then buyers push it down the chain.
- A consulting-led GHG inventory still costs thousands of dollars and weeks. SMEs do not have ERP feeds.
- They have a shoebox: electricity bills, diesel logs, flight CSVs, waste memos, commute surveys.
- If the SME cannot produce a cited inventory, the listed buyer cannot defend Scope 3.
- The gap is not another chatbot. It is turning messy source documents into numbers a reviewer can trace.

Six agents. Human in the loop. Not a chat window.

1 Ingestion

Gemini multimodal extract + verbatim excerpt

2 Review

Confirm, edit or reject before totals

3 Mapping

Select a curated factor — never invent one

4 Gaps

IFRS S2 / GHG Protocol holes, capped at six

5 Draft

Disclosure markdown with [A#] citations

6 Ask

Grounded assurance Q&A, cited answers

Fail-closed mapping + an append-only audit trail

- FastAPI agents in pipeline.py; orchestration, IDs, totals in orchestrator.py.
- React / Vite six-step UI: Upload → Review → Emissions → Gaps → Report → Ask.
- Factor library is bundled JSON (illustrative EPA/DEFRA-style). Unknown factor_id is dropped.
- Default activity status is pending. Rejected rows never enter totals.
- Claim is narrow: audit-grade draft — not a certified verification opinion.
- Offline Mock mode if no GEMINI_API_KEY; live Gemini when a key is set.

Why Gemini and Google Cloud

- Multimodal structured extraction: text, CSV, PDF and image bytes → JSON schema, not free prose.
- Constrained tool-use: the model may pick a factor_id that exists. Invented IDs never land in the inventory.
- Reproducible line: quantity × cited factor. Click [A#] to the source excerpt.
- Intended runtime: Cloud Run in asia-southeast1; Gemini key in Secret Manager.
- Honest status (6 Oct 2026): project greenledger-aibc is active but billing is off — no public URL yet.

GreenLeaf Trading Co. — verified sample pack

12

activity records

13.69

tCO₂e total

5–6

IFRS S2 gaps (on-screen)

S1/S2/S3

2.91 / 2.51 / 8.27 t

Gemini total is stable across 27 Sep (commit 2c654a8) and 6 Oct 2026 re-run.

Gap count drifted 6 → 5. In the video, read the on-screen gap count.

Mock mode on the same files: 14.00 tCO₂e / 4 canned gaps — do not mix modes.

Built for the firms the value chain actually runs on

- Who: SME finance and ops leads answering IFRS S2-style questionnaires without an ESG department.
- Demo scale: six synthetic source documents; minutes, not weeks.
- Unit of work is the source document. Citations and the audit trail are the product.
- Scale path: swap locale factor packs (SG / HK / JP / AU grid). Keep the pipeline.
- Commercial wedge: per-report SaaS; later white-label for financed-emissions evidence. The hackathon artifact is the working agent.

Load sample → extract → confirm → map → totals → gaps → cite → ask

- Load sample company (or swap electricity/gas PDFs for a multimodal take).
- Run Ingestion Agent. Confirm or correct rows. Mapping Agent cannot invent factors.
- Read 13.69 tCO₂e when the badge says Gemini. Then run gaps and open one [A#] chip.
- Ask: “Which records drive Scope 2?” — answer is grounded and cited (demo: A4).
- Every pipeline action is in the audit-trail drawer.

GreenLedger — recruiting to 2–4 by 11 October 2026

- Leader / backend & AI: FENG YIGEN (confirmed) — Hangzhou, JAPAC. Through the 4 Dec finale.
- Open: frontend / full-stack. Open: English narrative. Optional: ESG factor-library review.
- Platform rule: at least two members to submit. Waitlisted extra accounts cannot Accept yet.
- This deck does not list people who have not joined.

Disclosure-grade sustainability, without an ESG department

- Prototype URL: pending Cloud Run (billing / contest credits).
- Video: local MP4 via demo.storyboard.json — not uploaded until the captain posts YouTube/Vimeo/Drive.
- Proposal: docs/proposal.pdf · This deck: docs/greenledger-deck.pptx + .pdf
- Public GitHub: required by FAQ; no remote created yet.